THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · HARD LEVEL UNIT TEST SET 1

Chapters 1–4: **Coordinates, Linear Polynomials, World of Numbers, Algebraic Identities** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. The area of the triangle with vertices $(0,0)$, $(10,0)$ and $(0,14)$ is: 1

(a) 72 **(b)** 70 **(c)** 69 **(d)** 71

2. The area (in square units) of the triangle formed by the line $y = 2x + 10$ and the two axes is: 1

(a) 24 **(b)** 25 **(c)** 26 **(d)** 27

3. The value of $0.\overline{2} + 0.\overline{5}$ is: 1

(a) $\frac{7}{9}$ **(b)** $\frac{8}{9}$ **(c)** $\frac{7}{10}$ **(d)** $\frac{2}{3}$

4. If $x + y = 5$ and $xy = 6$, then $x^3 + y^3 =$ 1

(a) 35 **(b)** 36 **(c)** 34 **(d)** 37

5. The distance of the circumcentre of the right triangle with vertices $(0,0)$, $(6,0)$ and $(0,8)$ from the origin is: 1

(a) 7 **(b)** 6 **(c)** 5 **(d)** 4

6. The x-coordinate of the point where $y = 5x + 1$ and $y = 2x + 13$ meet is: 1

(a) 3 **(b)** 6 **(c)** 5 **(d)** 4

7. $0.\overline{13} + 0.\overline{31}$ equals: 1

(a) $\frac{5}{9}$ **(b)** $\frac{2}{5}$ **(c)** $\frac{1}{3}$ **(d)** $\frac{4}{9}$

8. If $x - y = 2$ and $xy = 15$, then $x^3 - y^3 =$ 1

(a) 100 **(b)** 98 **(c)** 97 **(d)** 99

9. A circle has diameter with endpoints $(1,-2)$ and $(19,78)$. Its radius is: 1

(a) 40 **(b)** 42 **(c)** 43 **(d)** 41

10. If $p(x) = 2x + 3$, then $p(p(4)) =$ 1

(a) 26 **(b)** 27 **(c)** 25 **(d)** 24

11. The smallest natural number n for which $\frac{n}{120}$ has a terminating decimal expansion is: 1

(a) 5 **(b)** 2 **(c)** 3 **(d)** 4

12. If $x + \frac{1}{x} = 3$, then $x^3 + \frac{1}{x^3} =$ 1

(a) 20 **(b)** 17 **(c)** 19 **(d)** 18

13. **Assertion (A):** The point $(-3, 4)$ lies in Quadrant II. 1
Reason (R): In Quadrant II the abscissa is negative and the ordinate is positive.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

14. **Assertion (A):** The zero of the polynomial $p(x) = 3x - 12$ is 4. 1
Reason (R): The zero of $ax + b$ $(a \neq 0)$ is $-\dfrac{b}{a}$.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

15. **Assertion (A):** The number 0.375 is a rational number. 1
Reason (R): Every terminating decimal can be written in the form $\dfrac{p}{q}$ with integers p, q and $q \neq 0$.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. The vertices of a right triangle are $O(0, 0)$, $A(6, 0)$ and $B(0, 8)$. Find the circumcentre and the circumradius. 2

17. Plan A costs ₹200 plus ₹30 per unit; Plan B costs ₹440 plus ₹10 per unit. For how many units do both cost the same? 2

18. Express $0.\overline{17} + 0.\overline{71}$ as a single fraction in lowest terms. 2

19. If $x + y = 8$ and $xy = 12$, find $x^3 + y^3$. 2

20. The end points of a diameter of a circle are $A(1, 1)$ and $B(13, 17)$. Find the centre and the radius. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. Find the points on the x-axis whose distance from $A(2, 20)$ is 25 units. 3

22. Find the point of intersection of $y = 5x + 2$ and $y = 2x + 11$. 3

23. Express $0.4\overline{7}$ as a fraction in lowest terms and verify by long division. 3

24. If $x + \frac{1}{x} = 7$, find $x^3 + \frac{1}{x^3}$. 3

OR

Three vertices of a rectangle $ABCD$ are $A(0,0)$, $B(9,12)$ and $D(-16,12)$. Find C and the diagonal AC. 3

25. A linear polynomial has $p(1)=12$ and $p(3)=20$. Find $p(x)$, its zero and $p(6)$. 3

OR

If $x=2+\sqrt{3}$, find $x^2+\frac{1}{x^2}$. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

26. **Case study.** On the map of a town drawn on a grid (1 unit = 1 km), Kavya's school is at A $(-3,4)$, a park is at B$(-3,-2)$ and the market is at C$(5,-2)$. 5

(i) In which quadrant does the school lie?

(ii) Write the coordinates of the reflection of the school in the x-axis.

(iii) Find the distance between the school and the park.

(iv) A bus stop is to be placed midway between the school A and the market C. Find its coordinates and the distance AC.

27. Find the decimal expansion of $\frac{3}{7}$ by long division, state the repeating block and its length, and hence write $\frac{3}{7}$ as a repeating decimal. 5

OR

If $x-y=3$ and $xy=28$, find x^2+y^2, x^3-y^3 and $(x+y)^2$. 5

END OF QUESTION PAPER Total: 50 Marks

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